

A SYSTEMATIC APPROACH TO THE RIEMANN-ROCH THEOREM

CLAUDIO BASTIANI-FONCK

ABSTRACT. In this paper, the reader will be introduced to the Riemann-Roch theorem for projective curves, as well as many interesting tools required to prove this theorem. The main goal of the paper is to provide an introductory glimpse into the extremely deep world of Riemann-Roch type theorems, as well as some of the methods and machinery used to prove the theorem.

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1. INTRODUCTION

The Riemann-Roch theorem has numerous formulations in many fields of mathematics. In a general sense, Riemann-Roch theorems relate some topological information of a space, like the genus, to algebraic information, such as the dimension of the vector space of functions with some constraints on their zeroes and poles. This type of idea turns out to be extremely important in many fields, including number theory, Riemannian geometry, and algebraic geometry. This paper is intended as an introduction to this style of theorem, with the goal of captivating the reader in the world of Riemann-Roch type theorems. The reader will be introduced to tools such as line bundles and cohomology, allowing for a systematic and motivated proof. The difficulty of reading this paper follows an approximately linear path as the goal is to motivate advanced machinery and then introduce it as it becomes helpful. In general proof sketches are given; however, technical details are often pushed to the bibliography.

2. THE LANGUAGE OF DIVISORS

In the following sections, I will prove one of the lower hanging results in the style of a Riemann-Roch theorem. This will illuminate the general flavour of these types

of theorems. I will assume that the reader has an understanding of the language of classical algebraic geometry, point set topology, and commutative algebra.

For the remainder F will denote a nonsingular projective plane curve and k an algebraically closed field.

Definition 2.1 (Divisors). We denote $\text{Div } F$ as the free abelian group generated by points of F . Therefore a *divisor* $D \in \text{Div } F$ is given by a formal linear combination of finitely many points of F

$$D = a_1 P_1 + \cdots + a_n P_n \quad P_i \in F \quad a_i \in \mathbb{Z}$$

with addition of divisors given pointwise. We call the coefficient in front of some $P \in F$ the *multiplicity* of D at P . The degree of a divisor is given by the formula

$$\deg D = a_1 + \cdots + a_n$$

which counts how many points the divisor has with multiplicity. We can define a partial order on divisors by $D_1 \geq D_2$ if at every point the multiplicity of D_1 is greater than or equal to the multiplicity of D_2 . We call a divisor *effective* if $D \geq 0$, i.e. it has nonnegative multiplicity at every point.

Construction 2.2 (Divisor of a Rational Function). Consider a rational function φ in the function field $K(F)$ of F . In this context the rational functions are exactly those functions which can be written as $\varphi = f/g$ for $f, g \in S_d(F)$, where $S_d(F)$ denotes the *homogeneous polynomials* of degree d . In words, the rational functions are quotients of homogeneous polynomials of the same degree.

Since F is assumed to be nonsingular, the local ring $\mathcal{O}_{F,P}$ at every point P is a discrete valuation ring. Note that f/g can be regarded as an element of the fraction field of $\mathcal{O}_{F,P}$ and therefore its valuation can be computed using the formula for the order of a polynomial h

$$\text{ord}_P(h) := \dim_k \mathcal{O}_{F,P} / \langle h \rangle \quad \mu_P(f/g) := \text{ord}_P(f) - \text{ord}_P(g)$$

which we will call the *multiplicity* of φ at P . Now we can define the divisor of φ as:

$$\text{div } \varphi = \sum_{P \in F} \mu_P(\varphi) \cdot P$$

The divisor of a rational function essentially encodes the information of its zeroes and poles. If φ has positive multiplicity at a point this means it has a zero of that degree at that point. Similarly, if φ has negative multiplicity at a point this means it has a pole of this degree at that point. Since the multiplicity at a point is a form of valuation, the following general results follow:

- (a) The valuation is a group homomorphism to $\mathbb{Z}$. This means that $\mu_P(\varphi_1 \cdot \varphi_2) = \mu_P(\varphi_1) + \mu_P(\varphi_2)$ which implies $\text{div}(\varphi_1 \cdot \varphi_2) = \text{div } \varphi_1 + \text{div } \varphi_2$.
- (b) The ultrametric inequality, which states that $\mu_P(\varphi_1 + \varphi_2) \geq \min(\mu_P(\varphi_1), \mu_P(\varphi_2))$ with a guarantee of equality when $\mu_P(\varphi_1) \neq \mu_P(\varphi_2)$.

This concept should remind the reader of multiplicities of zeroes and poles in complex analysis.

Remark 2.3. It is important to recognize that the definition of the multiplicity of some $f \in S_d(F)$ at a point can be rewritten as

$$\text{ord}_P f = \dim_k \mathcal{O}_{F,P} / \langle f \rangle = \dim_k \frac{\mathcal{O}_{\mathbb{P}^2,P} / \langle F \rangle}{\langle f \rangle} = \dim_k \mathcal{O}_{\mathbb{P}^2,P} / \langle F, f \rangle$$

which is exactly the intersection number $\mu_P(F, f)$ of F and f at P . This allows us to restate some foundational results about intersection numbers in the language of divisors.

Theorem 2.4 (Bezout's Theorem with Divisors). *Suppose that $f \in S(F)$ is homogeneous. Then:*

$$\deg \text{div } f = \deg \sum_P \mu_P(f, F) P = \deg f \cdot \deg F$$

Proof. This is a reformulation of Bezout's theorem in the language of divisors which can be found in [3]. $\square$

Theorem 2.5 (AF+BG Theorem with Divisors). *Suppose that $h, g \in S(F)$ are two nonzero homogeneous polynomials and $\text{div } h \geq \text{div } g$. Then there exists a homogeneous $b \in S(F)$ of degree $\deg h - \deg g$ with $h = bg$ in $S(F)$.*

Proof. This proof is found in [3]. To recap, since F is nonsingular its local ring at any point P is a discrete valuation ring. Noether's condition at P is then equivalent to $\mu_P(h, F) \geq \mu_P(g, F)$. For this to be satisfied at every point is just the statement that $\text{div } h \geq \text{div } g$. $\square$

Corollary 2.6. *If φ is a rational function then $\deg \text{div } \varphi = 0$.*

Proof. Note that $\varphi = f/g$ for some $f, g \in S(F)$ of the same degree. Then

$$\deg \text{div } \varphi = \deg \text{div } f - \deg \text{div } g = \deg f \cdot \deg F - \deg g \cdot \deg F = 0$$

by Theorem 2.4. $\square$

Definition 2.7 (Picard Group). The set of divisors originating as $\text{div } \varphi$ for some rational function φ form a subgroup of $\text{Div } F$ which we call $\text{PDiv } F$ or the principal divisors of F . The quotient

$$\text{Pic } F := \frac{\text{Div } F}{\text{PDiv } F}$$

is called the Picard group of F .

3. A CLASSICAL RESULT

We would now like to understand "how many" functions there are with prescribed zeroes and poles. This is the essence of the Riemann-Roch theorem.

Definition 3.1 (Space of Functions). Given a divisor D , we define

$$L(D) := \{\varphi \in K(F) : \text{div } \varphi + D \geq 0\} \cup \{0\}$$

so that if D has some positive multiplicity at a point P , then $\varphi \in L(D)$ is allowed to have a pole of at most that order on P . If D has some negative multiplicity at a point P , then φ is required to have a zero of at least that order on P .

By the ultrametric inequality $L(D)$ is a vector space and we can define its dimension:

$$l(D) = \dim_k L(D)$$

Using this definition we now have a goal. That is to compute and understand $l(D)$ for general divisors. This is what the Riemann-Roch theorem will accomplish.

Remark 3.2. If two divisors D_1 and D_2 are equivalent in $\text{Pic } F$ then they are related by a rational function $D_2 = D_1 + \text{div } \phi$. In this case the map

$$\psi : L(D_1) \rightarrow L(D_2), \varphi \mapsto \phi^{-1} \cdot \varphi$$

is an isomorphism with inverse $\varphi \mapsto \phi \cdot \varphi$.

Due to this the difference between divisors which are the same in the Picard group is uninteresting. This is why we define the Picard group. We are quotienting the differences we don't care about to focus on what is interesting.

Example 3.3. If $\deg D < 0$ then $l(D) = 0$.

Proof. Suppose for the sake of contradiction that $\varphi \in L(D)$. By Corollary 2.6 $\deg \text{div } \varphi = 0$ and

$$\deg(\text{div } \varphi + D) = \deg D < 0$$

which contradicts the assumption that $\text{div } \varphi + D \geq 0$. $\square$

Lemma 3.4. If D_1 and D_2 are two divisors with $D_1 \leq D_2$ then:

$$l(D_1) \leq l(D_2) \leq l(D_1) + \deg(D_2 - D_1)$$

Proof. It is clear that $l(D_1) \leq l(D_2)$. Therefore we only need to show that $l(D_2) \leq l(D_1) + \deg(D_2 - D_1)$. Since D_2 is given by adding some number of points to D_1 it suffices by induction on $\deg(D_2 - D_1)$ to show that

$$l(D + P) \leq l(D) + 1$$

for any point $P \in F$. Choose a rational function t that is a uniformizer of the local ring at P , suppose that D has multiplicity n at P , and consider the map:

$$\psi : L(D + P) \rightarrow k, \varphi \mapsto (t^{n+1}\varphi)(P)$$

This map is well defined since

$$\mu_P(t^{n+1}\varphi) = n + 1 + \mu_P(\varphi) \geq 0$$

because $D + P$ has multiplicity at most $n + 1$ at P . The kernel of the map is exactly those rational functions with $n + 1 + \mu_P(\varphi) \geq 1 \implies \mu_P(\varphi) \geq -n$, i.e. those in $L(D)$. Therefore we obtain one of the following exact sequences:

$$\begin{cases} 0 \rightarrow L(D) \xrightarrow{\text{incl}} L(D + P) \xrightarrow{\psi} 0, & \text{if } \text{im } \psi \text{ is trivial} \\ 0 \rightarrow L(D) \xrightarrow{\text{incl}} L(D + P) \xrightarrow{\psi} k \rightarrow 0, & \text{if } \text{im } \psi \text{ is nontrivial} \end{cases}$$

In the first case $l(D + P) = l(D) + 0 = l(D)$ and in the second $l(D + P) = l(D) + \dim k = l(D) + 1$. $\square$

Theorem 3.5.

$$l(D) \leq \max(0, \deg D + 1)$$

Proof. For some point P consider $D' = D - (n + 1)P$ where $n = \deg D$. Then by Example 3.3 $l(D') = 0$. Furthermore by Lemma 3.4

$$l(D) = l(D' + (n + 1)P) \leq 0 + n + 1 \leq n + 1$$

as desired. $\square$

In particular $l(0) = 1$ as the constant functions are a 1-dimensional space. This result may be thought of as an algebraic analogue of Liouville's theorem in complex analysis, which states that any holomorphic (global regular) function on the Riemann sphere ($\mathbb{P}^1$) is constant. While this classical result is very interesting, it can be strengthened to an equality.

Theorem 3.6 (Riemann-Roch). *For any line l not equal to F we claim that $K_F := (\deg F - 3) \operatorname{div} l$, called the canonical divisor, defines the same divisor class in the Picard group. Moreover if g is the genus of F then*

$$l(D) - l(K_F - D) = \deg D - g + 1$$

and $\deg K_F = 2g - 2$.

For now the reader should not worry about the exact definitions of K_F and g . In an informal sense K_F gives information about the structure of the differential forms and g gives information about the number of "handles" of F topologically.

While it is possible to prove this theorem in a very classical style, all these proofs turn out to be very miraculous. They seem to work via a coincidence of algebraic manipulation rather than a motivated reason. Therefore in order to give a more motivated proof we must introduce new machinery and move to a more general setting. Those who are interested in seeing a full classical proof are encouraged to read [3].

4. LINE BUNDLES

In this section I would like to talk about line bundles. In particular, I will show that on a smooth proper curve a line bundle carries the information of a divisor. In this way we can prove theorems about divisors by understanding line bundles. The reader is expected to be comfortable with the basic language of schemes and sheaves.

Definition 4.1 (Sheaf of Modules). A (pre)sheaf of modules on a scheme X is a (pre)sheaf which assigns to each open set $U \subseteq X$ an $\mathcal{O}_X(U)$ -module $\mathcal{F}(U)$. If $U_2 \subseteq U_1$ then $\mathcal{F}$ is compatible with the restriction map $\phi : U_1 \rightarrow U_2$ in the sense that if $f \in \mathcal{O}_X(U_1)$ and $m \in \mathcal{F}(U_1)$ then:

$$\phi^*(f \cdot m) = \phi^*(f) \cdot \phi^*(m)$$

A morphism $\psi : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ of (pre)sheaves of modules is a collection of $\mathcal{O}_X(U)$ -module morphisms $\psi_U : \mathcal{F}_1(U) \rightarrow \mathcal{F}_2(U)$ which are compatible with the restriction maps.

Example 4.2. Suppose that $X = \operatorname{Spec} R$ is an affine scheme and M is an R -module. Then the sheaf $\widetilde{M}$ associated to M is defined as:

$$\begin{aligned} \widetilde{M}(U) := \{(\varphi_{\mathfrak{p}} \in M_{\mathfrak{p}})_{\mathfrak{p} \in U} : \text{For every } \mathfrak{p} \in U \text{ there is an open neighborhood } U_{\mathfrak{p}} \text{ and} \\ m \in M, f \in R \text{ with } \varphi_{\mathfrak{q}} = m/f \text{ for each } \mathfrak{q} \in U_{\mathfrak{p}}\} \end{aligned}$$

Therefore an element of $\widetilde{M}(U)$ is a collection of $\varphi_{\mathfrak{p}} \in M_{\mathfrak{p}}$ so that each point has an affine open neighborhood where $\varphi_{\mathfrak{p}}$ is represented by the "same" element. The motivation for this definition is if M, N are R -modules then maps $M \rightarrow N$ are in bijective correspondence with maps $\widetilde{M} \rightarrow \widetilde{N}$. In this way there is an equivalence of categories between R -modules and associated sheaves of $\operatorname{Spec} R$ [1].

Definition 4.3 (Types of Sheaves of Modules). Suppose that $\mathfrak{p} \in X$. Then if we can always find a neighborhood U around $\mathfrak{p}$ satisfying the following hierarchical conditions $\mathcal{F}$ may also be one of the following:

Quasi-coherent: There is an $\mathcal{O}_X(U)$ -module M_U with $\mathcal{F}|_U \cong \widetilde{M_U}$.

Coherent: M_U is finitely generated over $\mathcal{O}_X(U)$.

Locally free: $M_U \cong \mathcal{O}_X(U)^{n_U}$ for some n_U .

Line bundle: $n_U = 1$ so that $M_U \cong \mathcal{O}_X(U)$.

By construction these are all local properties.

It is a theorem in [1] that quasi-coherence can be checked on any affine open cover. Furthermore if X is a Noetherian scheme then coherence can be checked on any affine open cover. In this paper we will be mainly concerned with line bundles.

For the remainder of the section C will denote an integral scheme of dimension 1, proper over k , all of whose local rings are regular. The sum of these quantifiers is to say that C is a proper smooth curve. The theory of these objects is developed in [1]. It is not strictly necessary for the reader to understand this, one may think of C as a curve as in the previous sections.

Example 4.4. Suppose that L is a line bundle over C . Since C is an irreducible topological space the open sets form a directed set with order given by inclusion. In particular for any two nonempty open sets U_1, U_2 there is the lower bound $U_1 \cap U_2$. Therefore take a filtered colimit $\varinjlim_U L(U)$ over all open sets of C . When $L = \mathcal{O}_C$ an element of the directed limit can be identified with a function that is regular on some nonempty open set, therefore this is the space of rational functions on C . This is called the *space of rational sections of L* or the stalk of L at the *generic point* of C and is denoted $L \otimes_{\mathcal{O}_C} K(C)$. Note that $L \otimes_{\mathcal{O}_C} K(C)$ is a $K(C)$ vector space.

Remark 4.5. Because C is a smooth proper curve there is a well defined notion of a divisor as a formal sum of points as in Definition 2.1 [1]. Furthermore given a rational function $\varphi \in K(C)$ there is a well defined notion of the divisor of φ as in Construction 2.2 [1].

Proof. A complete treatment of divisors of curves is found in [1]. Essentially an element $\varphi \in K(C)$ can be considered an element of the fraction field of the local ring $\mathcal{O}_{C,P}$ at a point. Since the local ring is a discrete valuation ring by assumption we can take the multiplicity at P to be the valuation $\mu_P(\varphi)$. $\square$

Construction 4.6 (Line Bundle Associated to a Divisor). For a divisor D we can form the *associated line bundle* as

$$\mathcal{L}^D(U) = \{f \in K(C) : \text{div } f|_U + D|_U \geq 0\}$$

where the notation $D|_U$ means that we are ignoring all points outside of U .

Proof. We need to show that $\mathcal{L}^D$ is a line bundle. Since C is a smooth curve all its points (excluding the generic point) are closed and it is a T_1 space with respect to these points. This will be used implicitly in the proof.

Suppose that $P \in C$. Let U be a neighborhood of P which does not contain any other point on which D has nonzero multiplicity. If D has 0 multiplicity on

P then we are done by definition. Otherwise let $n = \mu_P(D)$. Since C is a smooth curve the local ring $\mathcal{O}_{C,P}$ is a discrete valuation ring so let $t \in K(C)$ be a uniformizing parameter. We can choose another open neighborhood $U' \subseteq U$ where t is regular and has no other zeroes or poles in U' . In this neighborhood

$$\mathcal{L}^D(U') = \{f \in K(C) : \operatorname{div} f|_{U'} + nP \geq 0\}$$

and we can define the maps

$$\psi : \mathcal{L}^D(U') \rightarrow \mathcal{O}_C(U'), \varphi \mapsto t^n \varphi \quad \psi : \mathcal{O}_C(U') \rightarrow \mathcal{L}^D(U'), \varphi \mapsto t^{-n} \varphi$$

which are mutually inverse and therefore define an isomorphism $\mathcal{L}^D|_{U'} \cong \mathcal{O}_C|_{U'}$. $\square$

Intuitively $\mathcal{L}^D$ carries the information of the functions satisfying the divisor D in a local sense. One of the overarching themes in algebraic geometry is that forcing things to be local makes them easier to work with. This is another example of this principle.

Construction 4.7 (Divisor of Line Bundle). Suppose that L is a line bundle over C and $s \in L \otimes_{\mathcal{O}_C} K(C)$ is a nonzero rational section. At any point $P \in C$ we may choose an affine open U where there is an isomorphism $L|_U \cong \mathcal{O}_C|_U$. This identifies s with a rational function $\varphi \in K(C)$ locally around P . Therefore define $\mu_P(s) := \mu_P(\varphi)$. Since we can do this around any point we may define:

$$\operatorname{div} s := \sum_{P \in C} \mu_P(s) \cdot P$$

In order for this to be well defined, we need to prove that our choice of isomorphism $L|_U \cong \mathcal{O}_C|_U$ does not matter. In particular if $\psi_1 : L|_U \rightarrow \mathcal{O}_C|_U$ and $\psi_2 : L|_U \rightarrow \mathcal{O}_C|_U$ are two isomorphisms and $\varphi_i = \psi_i(s)$ then

$$\varphi_2 = \psi_2(s) = \psi_2 \psi_1^{-1} \psi_1(s) = \psi_2 \psi_1^{-1}(\varphi_1)$$

however $\psi_2 \psi_1^{-1}$ is an isomorphism $\mathcal{O}_C|_U \rightarrow \mathcal{O}_C|_U$. Therefore it is given by multiplication by a regular function $\phi \in \mathcal{O}_C|_U$ that is nowhere 0 on U . In particular on U :

$$\varphi_2 = \phi \varphi_1 \implies \operatorname{div} \varphi_2 = \operatorname{div} \phi + \operatorname{div} \varphi_1 \implies \mu_P(\varphi_1) = \mu_P(\varphi_2)$$

since ϕ does not have multiplicity at P by construction.

We would like to define the divisor of a line bundle as the divisor of one of its rational sections. Unfortunately, this is noncanonical since we are arbitrarily choosing a rational section. However since $L \otimes_{\mathcal{O}_C} K(C)$ is a 1-dimensional $K(C)$ vector space our choice is actually well defined up to multiplication by a rational function. This motivates the following theorem.

Theorem 4.8. *If we define $\operatorname{Pic}' C$ as the group of line bundles over C under the tensor product $\otimes_{\mathcal{O}_C}$ then the maps*

$$\psi : \operatorname{Pic} C \rightarrow \operatorname{Pic}' C, D \mapsto \mathcal{L}^D \quad \psi^{-1} : \operatorname{Pic}' C \rightarrow \operatorname{Pic} C, L \mapsto \operatorname{div} s$$

given in Construction 4.6 and Construction 4.7 are mutually inverse isomorphisms of abelian groups.

Proof. There are a few things to check:

- (a) The choice of s does not change the image in the Picard group.
- (b) L is isomorphic to $\mathcal{L}^{\operatorname{div} s}$.

- (c) Two equivalent divisors D_1 and D_2 define the same line bundle.
- (d) $\mathcal{L}^{D_1} \otimes_{\mathcal{O}_C} \mathcal{L}^{D_2} \cong \mathcal{L}^{D_1+D_2}$.

Our proofs will all rely heavily on the structure of $\mathcal{L} \otimes_{\mathcal{O}_C} K(C)$ as a 1-dimensional $K(C)$ vector space.

- (a) Note that s is a basis for the space of rational sections of L , any other rational section can be written uniquely as st for $t \in K(C)$. If we choose another section $s' = st$, then $s'/s = t$ is a rational function and $\text{div } t = \text{div } s' - \text{div } s$ is trivial in the Picard group.
- (b) As before note every rational function is of the form st and consider the map:

$$\mathcal{L} \otimes_{\mathcal{O}_C} K(C) \rightarrow K(C) \quad st \mapsto t$$

The condition that st is regular on some open set U is that $\text{div } s|_U + \text{div } t|_U \geq 0$. This is exactly the condition that $t \in \mathcal{L}^{\text{div } s}(U)$ so the map defines an isomorphism $L \cong \mathcal{L}^{\text{div } s}$.

- (c) Suppose that $D_1 - D_2 = \text{div } \phi$; then the map

$$\mathcal{L}^{D_1} \otimes K(C) \rightarrow \mathcal{L}^{D_2} \otimes K(C) \quad \varphi \mapsto \varphi \cdot \phi$$

gives an isomorphism $\mathcal{L}^{D_1} \cong \mathcal{L}^{D_2}$.

- (d) The rational sections of $\mathcal{L}^{D_1} \otimes_{\mathcal{O}_C} \mathcal{L}^{D_2}$ is a tensor product of 1-dimensional $K(C)$ vector spaces. Therefore for any $a \in \mathcal{L}^{D_1}$ and $b \in \mathcal{L}^{D_2}$ every rational function is of the form tab for some $t \in K(C)$. Furthermore we can assume that $\text{div } a = D_1$ and $\text{div } b = D_2$ by (b). The condition that tab is regular on some open set U is that $\text{div } t|_U + \text{div } a|_U + \text{div } b|_U \geq 0$. This is just the condition that $t \in \mathcal{L}^{D_1+D_2}(U)$ as desired.

□

In this way, the line bundles encode the exact same information as the Picard group of divisors. This makes them an extremely useful bookkeeping device for reasoning about divisors.

Construction 4.9 (Dual Bundle). Given a line bundle L over C we define the *dual bundle* as

$$L^\vee := \text{Hom}_{\mathcal{O}_C}(L, \mathcal{O}_C)$$

and claim that $L \otimes L^\vee \cong \mathcal{O}_C$. In particular $(\mathcal{L}^D)^\vee \cong \mathcal{L}^{-D}$.

Proof. On any open U where L is trivial the evaluation morphism $\text{ev} : L \otimes L^\vee \rightarrow \mathcal{O}_C$ is an isomorphism. This is because we can (noncanonically) identify $L|_U$ with $\mathcal{O}_C|_U$ and write:

$$\mathcal{O}_C|_U \otimes \text{Hom}_{\mathcal{O}_C|_U}(\mathcal{O}_C|_U, \mathcal{O}_C|_U) \cong \mathcal{O}_C|_U \otimes \mathcal{O}_C|_U \cong \mathcal{O}_C|_U$$

Since ev is a canonical isomorphism it extends to a global isomorphism as desired.

□

For this reason line bundles are often called *invertible sheaves* since they have "inverses" in the sense of the tensor product.

Theorem 4.10. Suppose that L is a line bundle over C and $D = a_1P_1 + \dots + a_nP_n$ is an effective divisor. Then there is an exact sequence

$$0 \rightarrow L \rightarrow L \otimes_{\mathcal{O}_C} \mathcal{L}^D \rightarrow \prod_i k(P_i)^{a_i} \rightarrow 0$$

where

$$k(P)(U) := \begin{cases} k, & \text{if } P \in U \\ *, & \text{if } P \notin U \end{cases}$$

denotes the skyscraper sheaf at P .

Proof. Firstly suppose that $L = \mathcal{L}^{D'}$ for some divisor D' and let b_i be the multiplicity of D' at P_i . Then we need to prove that

$$0 \rightarrow \mathcal{L}^{D'} \rightarrow \mathcal{L}^{D'+D} \rightarrow \prod_i k(P_i)^{a_i} \rightarrow 0$$

is exact. Since D is effective there is the obvious inclusion $\mathcal{L}^{D'} \rightarrow \mathcal{L}^{D'+D}$. Now we need a collection of maps $\psi_i : \mathcal{L}^{D'+D} \rightarrow k(P_i)^{a_i}$. If π is a uniformizer of $\mathcal{O}_{C,P_i}$ then

$$\frac{(\pi^{-a_i-b_i})}{(\pi^{-b_i})} \cong k^{a_i}$$

as a submodule of $\text{Frac}(\mathcal{O}_{C,P})$. Any rational function $\varphi \in K(C)$ can be considered as an element of $\text{Frac}(\mathcal{O}_{C,P})$ so let $i(\varphi)$ be its image in $\frac{(\pi^{-a_i-b_i})}{(\pi^{-b_i})}$. This is not well defined unless $\mu_P(\varphi) \geq -a_i - b_i$, however this condition is enforced by the divisor $D' + D$ so define the map:

$$\psi_U : \mathcal{L}^{D'+D}(U) \rightarrow k(P_i)^{a_i}(U) \quad \psi_U(\varphi) = \begin{cases} i(\varphi), & \text{if } P_i \in U \\ \text{trivial}, & \text{if } P_i \notin U \end{cases}$$

We can check exactness on the stalks [4]. If P is one of the P_i we have

$$0 \rightarrow \mathcal{L}_P^{bP} \rightarrow \mathcal{L}_P^{(a+b)P} \rightarrow \frac{(\pi^{-a-b})}{(\pi^{-b})} \rightarrow 0$$

however by definition $\mathcal{L}_P^{nP} = (\pi^{-n})$ and exactness follows. If P is not one of the P_i then we get

$$0 \rightarrow \mathcal{O}_{C,P} \rightarrow \mathcal{O}_{C,P} \rightarrow 0$$

which is exact. $\square$

Example 4.11. One of the simplest examples of line bundles is those over $\mathbb{P}^1$. Note that over $\mathbb{P}^1$ the rational function defined by a Möbius transformation can be chosen to have a pole and a zero of order 1 on any 2 points we choose. Therefore when we quotient by the rational divisors in the Picard group we lose all the information of where the poles and zeros of a rational function are. Due to this line bundles over $\mathbb{P}^1$ are characterized by their degree and we define the *twisting sheaf* as

$$\mathcal{O}_{\mathbb{P}^1}(n) := \mathcal{L}^{nP}$$

for any point P .

5. KÄHLER DIFFERENTIALS AND THE CANONICAL DIVISOR

In this section the canonical divisor will be defined and motivated using the machinery of Kähler differentials. For the remainder of this section let R be a k -algebra, M an R -module, and S a manifold.

Definition 5.1 (Derivations). A map $\delta : R \rightarrow M$ is called a derivation if it satisfies the conditions:

Additivity: $\delta(f + g) = \delta(f) + \delta(g)$ for all $f, g \in R$.

Leibniz Rule: $\delta(fg) = f\delta(g) + g\delta(f)$ for all $f, g \in R$.

k -linearity: $\delta(sf) = s\delta(f)$ for all $f \in R, s \in k$.

The space of derivations is denoted $\text{Der}_{R/k}(M)$. Note that there is an action of R on this space by

$$(r \cdot \delta)(a) = r \cdot \delta(a)$$

making $\text{Der}_{R/k}(M)$ an R -module.

In algebraic geometry rings should be thought of as the functions on a space. In the context of a manifolds class, one may think of R as the smooth functions on a manifold. Then one learns that a derivation at a point p is an operator which satisfies the Leibniz rule

$$\delta(fg) = f(p)\delta(g) + g(p)\delta(f)$$

and the set of all derivations is the tangent space $T_p S$ [9]. Thinking algebraically, take $M = R/\mathfrak{m} \cong k$ for some maximal ideal $\mathfrak{m}$. Here, $\mathfrak{m}$ corresponds to the ideal of smooth functions vanishing at a point p on the manifold and the map $R \rightarrow M$ is evaluation at p . Therefore the derivations of the map $R \rightarrow M$ are by definition the tangent space of the manifold at p . In the algebraic context, we take this as the definition and derive a lot of familiar results. Essentially, the map $R \rightarrow M$ corresponds to evaluation of some $f \in R$ at the closed point $\mathfrak{m} \in \text{Spec}(R)$. In this way $\text{Der}_{R/k}(M)$ corresponds to the tangent space of $\text{Spec}(R)$ at $\mathfrak{m}$.

Remark 5.2.

If $A \subseteq R$ then the condition that some $\delta \in \text{Der}_{R/k}(M)$ is A -linear is equivalent to $A \subseteq \ker(\delta)$.

Proof. Suppose that $a \in A$.

$\Rightarrow$: Note $\delta(1) = \delta(1 \cdot 1) = \delta(1) + \delta(1)$ so $\delta(1) = 0$. Now $\delta(a) = a\delta(1) = 0$.

$\Leftarrow$: Note $\delta(ar) = a\delta(r) + r\delta(a) = a\delta(r)$.

□

Therefore instead of requiring that δ be k -linear we could have required that δ kills k .

Definition 5.3 (Kähler Differentials). We define $\Omega_{R/k}$ to be the free R module generated by the formal symbols df modulo the relations

- $d(f + g) = df + dg$ for all $f, g \in R$;
- $d(fg) = f dg + g df$ for all $f, g \in R$;
- $ds = 0$ for all $s \in k$.

We equip $\Omega_{R/k}$ with a map $d : R \rightarrow \Omega_{R/k}, f \mapsto df$ called the *universal derivation*. One can check that this map is a derivation.

In the context of manifolds, one can think of $\Omega_{R/k}$ as the space of 1-forms. The universal derivation is analogous to the exterior derivative, which maps 0-forms, or functions, onto 1-forms. This idea will become much clearer with the following theorem.

Theorem 5.4 (Universal Property of $\Omega_{R/k}$). *Given a derivation $\delta : R \rightarrow M$ there is a unique R -module map $\phi : \Omega_{R/k} \rightarrow M$ that makes the diagram commute*

$$\begin{array}{ccc} R & \xrightarrow{\delta} & M \\ \downarrow d & \searrow \phi & \\ \Omega_{R/k} & & \end{array}$$

so that $\text{Der}_{R/k}(M) = \text{Hom}_A(\Omega_{R/k}, M)$.

Proof. Suppose that $f, g \in R$ and $s \in k$. Commutativity of the diagram requires that $\phi(df) = \delta(f)$ which determines the map uniquely. We need to show that all the relations in Definition 5.3 are satisfied.

- $\phi(d(f+g)) = \delta(f+g) = \delta(f) + \delta(g) = \phi(df) + \phi(dg)$;
- $\phi(d(fg)) = \delta(fg) = f\delta(g) + g\delta(f) = \phi(fdg + gdf)$;
- $\phi(ds) = \delta(s) = 0$.

□

We can express this result in the language of manifolds. Recall that when $M = R/\mathfrak{m} \cong k$ the derivations $\text{Der}_{A/k}$ correspond to the tangent space at $\mathfrak{m}$. In manifolds, the first order differential forms are global sections of the dual space of the tangent space. This means that at a point, an element of the tangent space $x \in T_p S$ corresponds to the map $\omega \mapsto \omega(x)$ which evaluates a differential form on x . This gives a correspondence between $T_p S$ and maps $T_p^* S \rightarrow k$. The equation $\text{Der}_{R/k}(M) = \text{Hom}(\Omega_{R/k}, M)$ expresses this fact very directly in the algebraic context.

Kähler differentials are very difficult to compute by hand in general, however there exist two powerful exact sequences which allow one to compute Kähler differentials for most "reasonable" rings.

Lemma 5.5 (First Exact Sequence). *If B is an R -algebra then there is the exact sequence*

$$\Omega_{R/k} \otimes_R B \rightarrow \Omega_{B/k} \rightarrow \Omega_{B/R} \rightarrow 0$$

of B -modules.

Proof. It suffices to show that the sequence

$$0 \rightarrow \text{Hom}_B(\Omega_{B/R}, M) \rightarrow \text{Hom}_B(\Omega_{B/k}, M) \rightarrow \text{Hom}_B(\Omega_{R/k} \otimes_R B, M)$$

is exact for every B -module M . Note that

$$\text{Hom}_B(\Omega_{R/k} \otimes_R B, M) \cong \text{Hom}_R(\Omega_{R/k}, M)$$

by adjointness of extension/restriction of scalars. Then by Theorem 5.4 this sequence corresponds to:

$$0 \rightarrow \text{Der}_{B/R}(M) \rightarrow \text{Der}_{B/k}(M) \rightarrow \text{Der}_{R/k}(M)$$

The map $\text{Der}_{B/k}(M) \rightarrow \text{Der}_{R/k}(M)$ is the pullback to R . The kernel are those maps which kill R , which is equivalent to being R -linear by Remark 5.2. Therefore the kernel is exactly $\text{Der}_{B/R}(M)$ as desired. □

Geometrically, this sequence relates the differential forms on the domain and codomain of a map. The algebra homomorphism $R \rightarrow B$ corresponds to the morphism $f : Y \rightarrow X$ where $Y = \operatorname{Spec} B$ and $X = \operatorname{Spec} R$. Then the terms in the exact sequence correspond to the following geometric objects:

- $\Omega_{R/k} \otimes_R B$ corresponds to the *pullback* $f^*\Omega_X$, representing forms on X transported to Y .
- $\Omega_{B/k}$ corresponds to Ω_Y , the intrinsic differential forms on Y .
- $\Omega_{B/R}$ corresponds to the *relative differentials* $\Omega_{Y/X}$.

The first map is the pullback of forms. The relative differentials $\Omega_{B/R}$ capture the "vertical" geometry of the map; they represent the differential forms along the fibers of f by measuring the infinitesimal changes in Y that vanish in the directions coming from X .

Lemma 5.6 (Second Exact Sequence). *If I is an ideal of R and $B = R/I$ then there is the exact sequence*

$$I/I^2 \rightarrow \Omega_{R/k} \otimes_R B \rightarrow \Omega_{B/k} \rightarrow 0$$

of B -modules.

Proof. It suffices to show that the sequence

$$0 \rightarrow \operatorname{Hom}_B(\Omega_{B/k}, M) \rightarrow \operatorname{Hom}_B(\Omega_{R/k} \otimes_R B, M) \rightarrow \operatorname{Hom}_B(I/I^2, M)$$

is exact for every B -module M . Note that

$$\operatorname{Hom}_B(\Omega_{R/k} \otimes_R B, M) \cong \operatorname{Hom}_R(\Omega_{R/k}, M) \quad \operatorname{Hom}_B(I/I^2, M) \cong \operatorname{Hom}_R(I, M)$$

by adjointness of extension/restriction of scalars and the universal property of quotients respectively. Now we have

$$0 \rightarrow \operatorname{Hom}_B(\Omega_{B/k}, M) \rightarrow \operatorname{Hom}_R(\Omega_{R/k}, M) \rightarrow \operatorname{Hom}_R(I, M)$$

which by Theorem 5.4 corresponds to:

$$0 \rightarrow \operatorname{Der}_{B/k}(M) \rightarrow \operatorname{Der}_{R/k}(M) \rightarrow \operatorname{Hom}_R(I, M)$$

Then for $i \in I$ and $f \in R$

$$\delta(if) = i\delta(f) + f\delta(i) = f\delta(i)$$

since I kills M (it is an R/I -module). Therefore the map $\operatorname{Der}_{R/k}(M) \rightarrow \operatorname{Hom}_R(I, M)$ by restriction to I is well defined. By comparing definitions we see that an element of $\operatorname{Der}_{B/k}$ is an element of $\operatorname{Der}_{A/k}$ (via precomposition with the quotient map) that kills I . This is exactly the kernel of the map $\operatorname{Der}_{R/k}(M) \rightarrow \operatorname{Hom}_R(I, M)$ as desired. $\square$

Geometrically, this sequence relates the cotangent bundle of a submanifold $Y = \operatorname{Spec} B$ to its ambient space $X = \operatorname{Spec} R$. Then the terms in the exact sequence correspond to the following geometric objects:

- $\Omega_{R/k} \otimes_R B$ corresponds to $T^*X|_Y$, the cotangent bundle of the ambient space restricted to the submanifold.
- $\Omega_{B/k}$ corresponds to T^*Y , the intrinsic cotangent bundle of the submanifold.
- I/I^2 corresponds to the *conormal bundle* N^*Y .

The sequence states that the differential forms on Y are obtained by taking the ambient differential forms restricted to Y and "modding out" by the conormal directions (the differentials of the constraint equations defining Y). In the smooth manifold context, this map is injective on the left, reflecting the decomposition of the ambient cotangent space into tangent and normal components.

Example 5.7 (Cotangent Space). Suppose that $\mathfrak{m}$ is the kernel of a map $R \rightarrow k$. By applying Lemma 5.6 to $R/\mathfrak{m}$ we obtain the exact sequence

$$\mathfrak{m}/\mathfrak{m}^2 \rightarrow \Omega_{R/k} \otimes_R (R/\mathfrak{m}) \rightarrow 0$$

and see that $\mathfrak{m}/\mathfrak{m}^2 \cong \Omega_{R/k} \otimes_R k$.

Example 5.7 says that $\mathfrak{m}/\mathfrak{m}^2$ is the cotangent space of R at $\mathfrak{m}$. As k -vector spaces we can write $R \cong \mathfrak{m} \oplus k$. A derivation of the map $R \rightarrow k$ sends k to 0 and is determined by its restriction to $\mathfrak{m}$. However it must also send $\mathfrak{m}^2$ to 0 by the Leibniz rule. Therefore the derivations are in correspondence with k -linear forms on $\mathfrak{m}/\mathfrak{m}^2$. Then $\mathfrak{m}/\mathfrak{m}^2$ is dual to the tangent space at $\mathfrak{m}$, validating our result.

This is important for the following reason. We would like to say that $\text{Spec } R$ is *smooth* if the tangent space at every point has the same dimension as the Krull dimension of R . Since $\Omega_{R/k}$ is locally dual to the tangent space we can consider instead when it has this property. Indeed, Example 5.7 shows that R must be a *regular ring* in order for $\text{Spec } R$ to be smooth. Through some work it is possible to show that if R is a k -algebra of finite type then $\text{Spec } R$ is smooth if and only if R is a regular ring. This gives a nice algebraic characterization of this property.

Remark 5.8. Suppose that $\text{Spec } R$ is a regular one dimensional affine variety. Pick any nonzero differential $\omega \in \Omega_{R/k}$. At a point $\mathfrak{p} \in R$ the local ring $R_{\mathfrak{p}}$ is a discrete valuation ring so we can choose a uniformizer t . Then at the stalk ω can be written uniquely as $f dt$ for some $f \in R$. We can associate ω to the function f on some neighborhood U of $\mathfrak{p}$ and define $\mu_{\mathfrak{p}}(\omega) = \mu_{\mathfrak{p}}(f)$. Since we can define the multiplicity of ω locally around every point, we can define its divisor using this notion. It turns out that any differential form $\omega \in \Omega_{R/k}$ gives the same divisor class in the Picard group. This divisor class is therefore called the *canonical divisor*. An algebraic proof of this statement is hard, so we will instead approach it via the machinery of line bundles.

In the language of manifolds, a top degree differential form $f dx_1 \wedge \cdots \wedge dx_n$ can be associated to the function f . However this is not canonical since we need to choose the basis $x_1 \wedge \cdots \wedge x_n$. Furthermore this choice only associates ω to f on a local coordinate chart. Despite this, we can use this local notion to associate ω to a well defined divisor, and under reasonable assumptions our choices of coordinates do not change the divisor in the Picard group.

As the reader may have guessed, the "better" definition of the canonical divisor is to use $\Omega_{R/k}$ to define a line bundle. Then we can take the divisor associated to the line bundle. If we trace back definitions this is actually equivalent to our naive definition, however the machinery of line bundles allow us to avoid algebraic technicalities.

Lemma 5.9. *Consider the map*

$$\delta : R \otimes_k R \rightarrow R \quad f \otimes g \mapsto fg$$

and set $J := \ker \delta$. Then $J/J^2 \cong \Omega_{R/k}$.

Proof. Consider $R \otimes_k R$ as an R -module via left multiplication. There are the isomorphisms

$$\begin{aligned} J/J^2 &\rightarrow \Omega_R & f_i \otimes g_i &\mapsto f_i dg_i \\ \Omega_R &\rightarrow J/J^2 & df &\mapsto 1 \otimes f - f \otimes 1 \end{aligned}$$

which are inverses of each other. The details are worked out in [4, 15.4]. $\square$

Lemma 5.10 (Ideal Sheaf). *Suppose that X is a separated scheme and $Y \subseteq X$ is closed. The inclusion $i : Y \rightarrow X$ induces the following constructions:*

- The pushforward sheaf $(i_* \mathcal{O}_Y)(U) = \mathcal{O}_Y(i^{-1}(U))$ on X .
- The morphism of sheaves

$$i^\# : \mathcal{O}_X \rightarrow i_* \mathcal{O}_Y \quad \varphi \mapsto i^* \varphi$$

induced by the pullbacks $i_U^* : \mathcal{O}_X(U) \rightarrow \mathcal{O}_Y(i^{-1}(U))$.

- The ideal sheaf $\mathcal{I}_{Y/X}(U) = \ker i_U^*$ which is the kernel of $i^\#$.
- The exact sequence $0 \rightarrow \mathcal{I}_{Y/X} \rightarrow \mathcal{O}_X \rightarrow i_* \mathcal{O}_Y \rightarrow 0$.

Proof. Since all the constructions used preserve the sheaf property [4] we can check exactness locally. In particular we can assume that $X = \operatorname{Spec} R$ is affine. In this case $\mathcal{O}_X$ and $i_* \mathcal{O}_Y$ are the sheaves associated to the R -modules R and R/J , where $J := \mathcal{I}(Y)$. Therefore this sequence corresponds to the exact sequence:

$$0 \rightarrow J \rightarrow R \rightarrow R/J \rightarrow 0$$

$\square$

Construction 5.11 (Cotangent Sheaf). Suppose that X is a separated scheme. Therefore X is isomorphic to the diagonal Δ_X inside of $X \times X$. Then the map $i : X \rightarrow X \times X, x \mapsto x \times x$ has pullback δ as given in Lemma 5.9. Define $\mathcal{I}$ as the ideal sheaf of the diagonal of X . Now take

$$\Omega_X = i^*(\mathcal{I}/\mathcal{I}^2)$$

to be the pullback sheaf of $\mathcal{I}/\mathcal{I}^2$. On an affine open the pullback just considers J/J^2 as an R -module instead of an $R \otimes_k R$ -module [4, 15.5]. Therefore by Lemma 5.9

$$\Omega_X|_U = \tilde{\Omega}_{R/k}$$

on some affine open subset $U \cong \operatorname{Spec} R$.

Now we work towards a result which validates our intuition about differential forms. For this we must make an algebraic argument.

Proposition 5.12. *If X is a smooth variety of pure dimension n then Ω_X is locally free of rank n .*

Proof. By assumption the stalk $\Omega_{X,P}$ is a free $\mathcal{O}_{X,P}$ module generated by $ds_1, \dots, ds_n$. Therefore it suffices to show that for any finitely presented R -module M , if $\tilde{M}$ is stalkwise free then it is locally free.

Let $s_1, \dots, s_n \in M_{\mathfrak{p}}$ be a basis of $M_{\mathfrak{p}}$. If we write $s_i = f_i/g_i$ for $g_i \notin \mathfrak{p}$ and let

$g = g_1 \cdots g_n$ then by replacing M with the localization of M at g we can consider $s_i \in M$. In particular we can write:

$$\left(\frac{M}{\langle s_1, \dots, s_n \rangle} \right)_{\mathfrak{p}} = \frac{M_{\mathfrak{p}}}{\langle s_1, \dots, s_n \rangle_{\mathfrak{p}}} = 0$$

Let $m_1, \dots, m_r$ be the generators for M . The condition that the map $\langle s_1, \dots, s_n \rangle_{\mathfrak{p}} \rightarrow M_{\mathfrak{p}}$ is a surjection implies that we can restrict to some open set $D(h)$ in the direct limit where $m_i \in \langle s_1, \dots, s_n \rangle_h$. By doing this for all m_i we assume that $M = \langle s_1, \dots, s_n \rangle$. Since M is finitely presented we may write

$$R^m \rightarrow R^n \rightarrow M \rightarrow 0$$

where the map $R_{\mathfrak{p}}^n \rightarrow M_{\mathfrak{p}}, r_i \mapsto s_i$ is an isomorphism. Since M is finitely presented let $t_1, \dots, t_m \in R^n$ be generators of the kernel of the map $R^n \rightarrow M$. The condition that $t_i = 0$ in $R_{\mathfrak{p}}^n$ means that there is some $l_1, \dots, l_m \in R \setminus \mathfrak{p}$ with $l_i t_i = 0$. If we localize at the product of all the l_i then all the relations become trivial and we are done. $\square$

Essentially, we have proven that being free on stalks is equivalent to being free on an open cover (for finitely presented associated sheaves). This is some kind of "local to less local" principle. The following theorem may be interesting to those who want to explore more of commutative algebra ideas behind these statements:

Theorem 5.13. *The following are equivalent*

- (a) M is finite and $\text{rk}_M(\mathfrak{p}) := \dim_{\kappa(\mathfrak{p})}(M \otimes_R \kappa(\mathfrak{p}))$ (where $\kappa(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}$ is the residue field) is locally constant in the Zariski topology;
- (b) M is finitely presented and for all maximal ideals $M_{\mathfrak{m}}$ is free;
- (c) M is finitely presented and for all prime ideals $M_{\mathfrak{p}}$ is free;
- (d) M is finitely presented and R -flat;
- (e) M is finitely projective.

Proof. The proofs of these statements may be found on the [stacks project](#). $\square$

Now that we have Proposition 5.12 we can finally give a rigorous definition of the canonical divisor.

Construction 5.14. (Canonical Bundle) Note that if $R_{\mathfrak{p}}$ is a discrete valuation ring with uniformizer t then $\Omega_{R_{\mathfrak{p}}/k}$ is the free $R_{\mathfrak{p}}$ -module generated by dt . Therefore if C is a smooth proper curve (so that all its local rings are discrete valuation rings) each stalk of the cotangent sheaf in Construction 5.11 is free of rank 1. Therefore by Proposition 5.12 Ω_C is a line bundle called the *canonical bundle*, and hence defines a unique divisor class in the Picard group called the *canonical divisor*.

The reader is encouraged to try to unravel the definitions and see why this definition of the canonical divisor is the same as the one given in Remark 5.8.

6. COHOMOLOGY OF SHEAVES

In this section the tool of Čech Cohomology will be developed and motivated. This will prove extremely useful in giving a sophisticated and systematic proof of the Riemann-Roch theorem.

Definition 6.1 (Čech Complex). Suppose that $U_1, \dots, U_r$ is an affine open cover of X and $\mathcal{F}$ is a sheaf on X . The *Čech complex* with respect to this cover is

$$C^p(\mathcal{F}) := \bigoplus_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_p})|_{U_{i_0} \cap \dots \cap U_{i_p}}$$

so that an element $\varphi \in C^p(\mathcal{F})$ is a collection of sections $\varphi_{i_0 < \dots < i_p} \in \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_p})$ on each intersection of $p+1$ open sets in the cover. We impose no restriction on the sections themselves. There is a *boundary map* $d_p : C^p(\mathcal{F}) \rightarrow C^{p+1}(\mathcal{F})$ defined by:

$$(d_p \varphi)_{i_0, \dots, i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \varphi_{i_0, \dots, \hat{i}_k, \dots, i_{p+1}}|_{U_{i_0} \cap \dots \cap U_{i_{p+1}}}$$

Definition 6.2 (Čech Cohomology). It can be verified via a combinatorial argument that $d^{p+1} \circ d^p = 0$ [1]. Therefore we form the *cochain complex*

$$C^0(\mathcal{F}) \xrightarrow{d^0} C^1(\mathcal{F}) \xrightarrow{d^1} C^2(\mathcal{F}) \xrightarrow{d^2} \dots$$

and define

$$H^p(X, \mathcal{F}) := \frac{\ker d^p}{\operatorname{im} d^{p-1}} \quad h^p(X, \mathcal{F}) := \dim_k H^p(X, \mathcal{F})$$

where $H^p(X, \mathcal{F})$ is the p -th cohomology of $\mathcal{F}$.

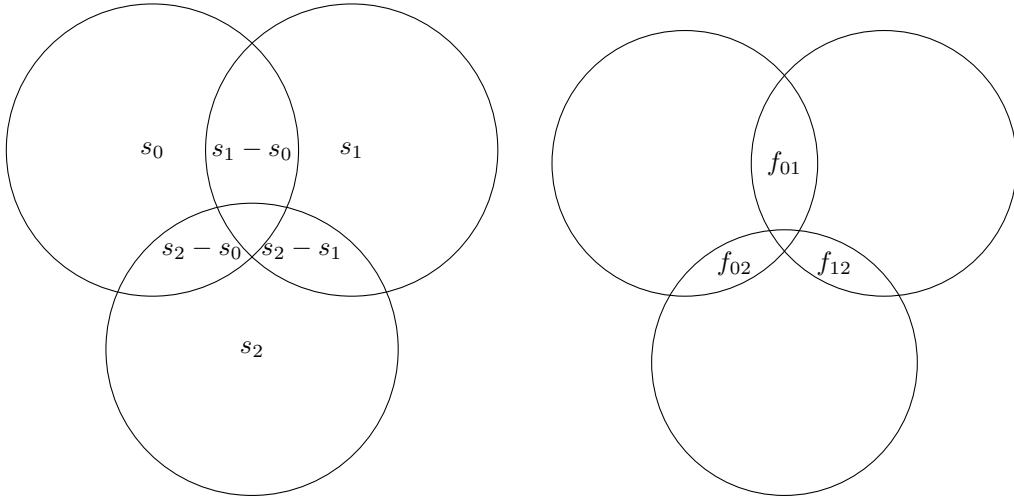
The Čech Cohomology essentially measures "how much" a collection of global sections can disagree. To see this we will discuss H^0 and H^1 . Firstly $H^0(X, \mathcal{F})$ is just the kernel of d^0 . Therefore write

$$(d^0 f)_{i_0, i_1} = f_{i_1}|_{U_{i_0} \cap U_{i_1}} - f_{i_0}|_{U_{i_0} \cap U_{i_1}} = 0 \implies f_{i_0}|_{U_{i_0} \cap U_{i_1}} = f_{i_1}|_{U_{i_0} \cap U_{i_1}}$$

which is exactly the sheaf condition that the $(f)_{i_1, i_2}$ are compatible on overlaps. In this way $H^0(X, \mathcal{F})$ corresponds to the *global sections* of $\mathcal{F}$, or those local sections which agree to "0th order". Now consider $H^1(X, \mathcal{F})$. A cocycle of $H^1(X, \mathcal{F})$ satisfies

$$(d^0 f)_{i_0, i_1, i_2} = (f_{i_1, i_2} - f_{i_0, i_2} + f_{i_0, i_1})|_{U_{i_0} \cap U_{i_1} \cap U_{i_2}} = 0$$

and a coboundary of $H^1(X, \mathcal{F})$ is the image of something in $C^1(X, \mathcal{F})$. This can be visualized as a Venn diagram:



We can consider the $s_i - s_j$ to be the "disagreements" of the sections s_i on the overlaps. The statement $d^1 \circ d^0(s_i) = 0$ says that the disagreement of s_0 and s_2 is the sum of the disagreements between s_0 and s_1 , and s_1 and s_2 . This can be viewed as a "first order" agreement condition. The cocycle condition means that we pick some f_{ij} on the intersections with this agreement property. However, we are not interested in f_{ij} which arise from some sections s_i . Therefore the homology asks, if the f_{ij} satisfy the first order agreement property, how might we be unable to glue them together into a system of sections s_i ? In the same way $H^p(X, \mathcal{F})$ asks, in what way might we fail to glue together sections of p -fold intersections satisfying a p th degree agreement condition (that is, a condition on $(p+1)$ -fold intersections) into a system of sections on the $(p-1)$ -fold intersections.

Theorem 6.3 (Long Exact Cohomology Sequence). *Suppose that*

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

is an exact sequence of chain complexes. Then there is a long exact sequence of cohomology:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(A) & & H^1(A) & & H^2(A) & \longrightarrow & \cdots \\ & & \downarrow & \nearrow & \downarrow & \nearrow & \downarrow & \nearrow & \\ & & H^0(B) & & H^1(B) & & H^2(B) & & \cdots \\ & & \downarrow & \nearrow & \downarrow & \nearrow & \downarrow & \nearrow & \\ & & H^0(C) & & H^1(C) & & H^2(C) & & \cdots \end{array}$$

Proof. We can visualize a short exact sequence of chain complexes in the following way:

$$\begin{array}{ccccccc} 0 & & 0 & & 0 & & 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ A^0 & \longrightarrow & A^1 & \longrightarrow & A^2 & \longrightarrow & A^3 & \longrightarrow & \cdots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ B^0 & \longrightarrow & B^1 & \longrightarrow & B^2 & \longrightarrow & B^3 & \longrightarrow & \cdots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ C^0 & \longrightarrow & C^1 & \longrightarrow & C^2 & \longrightarrow & C^3 & \longrightarrow & \cdots \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & & 0 & & 0 & & 0 & & \end{array}$$

The diagram is made up of maps of exact sequences of the form

$$\begin{array}{ccccccc} 0 & \longrightarrow & A^p & \longrightarrow & B^p & \longrightarrow & C^p & \longrightarrow & 0 \\ & & \downarrow d_A^p & & \downarrow d_B^p & & \downarrow d_C^p & & \\ 0 & \longrightarrow & A^{p+1} & \longrightarrow & B^{p+1} & \longrightarrow & C^{p+1} & \longrightarrow & 0 \end{array}$$

and applying the snake lemma gives the exact sequence:

$$0 \rightarrow Z^p A \rightarrow Z^p B \rightarrow Z^p C \rightarrow \frac{A^{p+1}}{dA^p} \rightarrow \frac{B^{p+1}}{dB^p} \rightarrow \frac{C^{p+1}}{dC^p} \rightarrow 0$$

where the Z^p denotes the p th cocycles (the notation of [8]). We can patch together the first half of this sequence for $p-1$ with the second half of the sequence for $p+1$ to obtain:

$$\begin{array}{ccccccc} \frac{A^p}{dA^{p-1}} & \longrightarrow & \frac{B^p}{dB^{p-1}} & \longrightarrow & \frac{C^p}{dC^{p-1}} & \longrightarrow & 0 \\ \downarrow d_A^p & & \downarrow d_B^p & & \downarrow d_C^p & & \\ 0 & \longrightarrow & Z^{p+1}A & \longrightarrow & Z^{p+1}B & \longrightarrow & Z^{p+1}C \end{array}$$

Applying the snake lemma again gives

$$H^p(A) \rightarrow H^p(B) \rightarrow H^p(C) \rightarrow H^{p+1}(A) \rightarrow H^{p+1}(B) \rightarrow H^{p+1}(C)$$

and the proposition follows from patching these sequences together for all p . $\square$

Theorem 6.4. *Let X be a separated scheme and $\mathcal{U} = \{U_i\}$ be an affine open cover. Let*

$$0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$$

be a short exact sequence of quasi-coherent sheaves. Then the induced sequence

$$0 \rightarrow C(\mathcal{F}_1) \rightarrow C(\mathcal{F}_2) \rightarrow C(\mathcal{F}_3) \rightarrow 0$$

is a short exact sequence of cochain complexes.

Proof. Because X is separated, every finite intersection $U_{i_0 \dots i_p}$ is affine. Restricting quasi-coherent sheaves to an affine open keeps them quasi-coherent. On an affine scheme $W = \text{Spec}(A)$, the global sections functor $\Gamma(W, -)$ is exact on quasi-coherent sheaves (equivalently $\text{QCoh}(W) \simeq A\text{-Mod}$), so for each $U_{i_0 \dots i_p}$ we have a short exact sequence:

$$0 \rightarrow \mathcal{F}_1(U_{i_0 \dots i_p}) \rightarrow \mathcal{F}_2(U_{i_0 \dots i_p}) \rightarrow \mathcal{F}_3(U_{i_0 \dots i_p}) \rightarrow 0.$$

However by definition the sequence

$$0 \rightarrow C(\mathcal{F}_1) \rightarrow C(\mathcal{F}_2) \rightarrow C(\mathcal{F}_3) \rightarrow 0$$

is a direct sum of these sequences, hence exact. $\square$

Now we will relate the Čech cohomology to another more canonical construction.

Definition 6.5 (Sheaf Cohomology). The category of sheaves of modules has *enough injectives*. Therefore the sheaf cohomology can be defined as the right derived functor to the global sections functor $\Gamma(X, -)$. More specifically if $\mathcal{F}$ is a sheaf and

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^0 \rightarrow \mathcal{I}^1 \rightarrow \dots$$

is an *injective resolution* then the sequence of global sections

$$0 \rightarrow \Gamma(X, \mathcal{I}^0) \rightarrow \Gamma(X, \mathcal{I}^1) \rightarrow \Gamma(X, \mathcal{I}^2) \rightarrow \dots$$

is a cochain complex and cohomology can be defined.

The right derived functor of a left exact functor like $\Gamma(X, -)$ essentially measures how much it fails to be right exact. The reader who is unfamiliar with the notion of a derived functor is encouraged to read that section in [8]. Alternatively, one may take results of Theorem 6.6 and Theorem 6.9 as a reasonable black box.

Theorem 6.6. *If X is a separated scheme and $\mathcal{F}$ is a quasi-coherent sheaf on X , then the Čech cohomology of $\mathcal{F}$, computed with respect to any affine open cover, is isomorphic to the sheaf cohomology.*

Proof. The proof is found in [1]. $\square$

Theorem 6.6 is the workhorse result of this section. Essentially, the sheaf cohomology is a very canonical construction and has a lot of nice theoretical properties. On the other hand it is very difficult to actually compute things with. The Čech cohomology is more explicit and allows one to compute by taking a "good" open cover. Therefore a parallel treatment allows one to prove a lot of nontrivial theorems by bouncing these cohomologies off each other.

Corollary 6.7. *If X is an affine scheme and $\mathcal{F}$ a quasi-coherent sheaf then $H^p(X, \mathcal{F}) = \{0\}$ for every $p \geq 1$.*

Proof. We may compute Čech cohomology by Theorem 6.6. Choose the open cover with X as the only set. The theorem follows from the fact that there is no way to choose more than one open set from a single open set. $\square$

Definition 6.8 (Flasque Sheaves). A sheaf $\mathcal{F}$ is *flasque* (or *flabby*) if the restriction maps $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ are surjective for all open sets $V \subset U$.

Theorem 6.9. *There are the following results on flasque sheaves:*

- (a) *All injective objects in the category of sheaves on X are flasque.*
- (b) *The global sections functor is exact on flasque sheaves.*
- (c) *For any flasque sheaf $\mathcal{F}$ we have $H^p(X, \mathcal{F}) = 0$ for all $p > 0$.*
- (d) *The sheaf cohomology may be computed with any flasque resolution. This is a relaxation of the condition requiring an injective resolution.*
- (e) *Flasque sheaves are preserved by direct limits.*

Proof. The proof of all these statements may be found in [1]. I will give a proof of (b) since it is equivalent to (c) and (d) by general results on derived functors [8]. Furthermore (a) is not necessary for this paper and (e) is a formal verification which can be found in [1] for those who are interested.

Suppose that

$$0 \rightarrow \mathcal{F}' \xrightarrow{a} \mathcal{F} \xrightarrow{b} \mathcal{F}'' \rightarrow 0$$

with $\mathcal{F}'$ a flasque sheaf. Since the global sections functor is left exact [4] it suffices to show that $\Gamma(U, b)$ is surjective for every open U . Suppose that $s \in \mathcal{F}''(U)$ and consider the set of pairs (V, t) for $V \subseteq U$ open and $t \in \mathcal{F}(V)$ with $b(t)|_V = s|_V$. By exactness of the sequence this set is not empty, and furthermore by Zorn's lemma has an upper bound (V, t) . For any $x \in U$ pick a neighborhood $W \subseteq U$ and a local lift $t' \in \mathcal{F}(W)$. Note that $t|_{W \cap V} - t'|_{W \cap V}$ lies in $a(\mathcal{F}'(W \cap V))$ after possibly shrinking W . Using flasqueness this difference extends to some $r \in \mathcal{F}'(W)$, therefore replace t' with $t' - a(r)$ so that t and t' agree on $W \cap V$. Now they can be glued into a global section on $U \cup V$ and $x \in V$. This shows that $V = U$ as desired. $\square$

Theorem 6.10 (Grothendieck Vanishing). *Suppose that X is Noetherian of dimension n and $\mathcal{F}$ is a quasi-coherent sheaf. Then $H^p(X, \mathcal{F}) = 0$ for all $p > n$.*

Proof. The full proof is in [1]. We give a proof sketch, and also implicitly assume that direct limits of sheaves commute with cohomology for Noetherian topological spaces. First we need a general construction. Suppose that $U \subset X$ is open and $Y = X - U$ is its complement. Now construct:

- The direct image sheaf of the inclusion $Y \rightarrow X$ along $\mathcal{F}|_Y$:

$$\mathcal{F}_Y(V) = \mathcal{F}(Y \cap V)$$

- The sheafification of the extension of $\mathcal{F}|_U$ by zero:

$$\text{sheafification of } \mathcal{F}_U(V) = \begin{cases} \mathcal{F}(V), & V \subseteq U \\ \{0\}, & \text{otherwise} \end{cases}$$

If we take stalks we see that

$$\mathcal{F}_{U,P} = \begin{cases} \mathcal{F}, & P \in U \\ \{0\}, & P \notin U \end{cases} \quad \mathcal{F}_{Y,P} = \begin{cases} \mathcal{F}, & P \in Y \\ \{0\}, & P \notin Y \end{cases}$$

therefore there is an exact sequence

$$0 \rightarrow \mathcal{F}_U \rightarrow \mathcal{F} \rightarrow \mathcal{F}_Y \rightarrow 0$$

which will be used repeatedly in this proof.

Now suppose that X is reducible. If Y is an irreducible closed subset and $U = X - Y$ we obtain

$$0 \rightarrow \mathcal{F}_U \rightarrow \mathcal{F} \rightarrow \mathcal{F}_Y \rightarrow 0$$

and by taking the long exact sequence of cohomology it suffices to show that $H^p(U, \mathcal{F}|_U) = \{0\}$ and $H^p(U, \mathcal{F}|_Y) = \{0\}$. Since X is Noetherian this process must terminate and it suffices to consider when X is irreducible. Now induct on the dimension n of X .

Base Case $n=0$: If X is irreducible of dimension 0 then the only open subsets of X are X and the empty set. Therefore X must be affine and we apply Corollary 6.7.

Inductive Step: We first perform a lot of reductions. Denote $B = \bigcup_{U \subseteq X} \mathcal{F}(U)$ and let A be the set of all finite subsets of B . For any $\alpha \in A$ we denote $\mathcal{F}_\alpha$ as the subsheaf of $\mathcal{F}$ generated by sections in α . Then A becomes a directed set and $\mathcal{F} = \varinjlim \mathcal{F}_\alpha$ so it suffices to prove each $\mathcal{F}_\alpha$ vanishes. For some subset $\alpha' \subseteq \alpha$ there is an exact sequence

$$0 \rightarrow \mathcal{F}_{\alpha'} \rightarrow \mathcal{F}_\alpha \rightarrow \mathcal{G} \rightarrow 0$$

where $\mathcal{G}$ is a sheaf generated by $\#(\alpha - \alpha')$ sections. By taking the long exact sequence of cohomology it suffices to prove that $H^p(X, \mathcal{F}'_\alpha) = H^p(X, \mathcal{G}) = \{0\}$ for $p > n$. Therefore by induction we reduce to the case where $\mathcal{F}$ is generated by a single section over an open set U . In this case $\mathcal{F}$ is a quotient of the constant sheaf Z_U . If $\mathcal{R}$ is the kernel sheaf then by taking the long exact sequence of cohomology on

$$0 \rightarrow \mathcal{R} \rightarrow Z_U \rightarrow \mathcal{F} \rightarrow 0$$

it suffices to prove the result on $\mathcal{R}$ and Z_U . For $x \in U$ stalk $\mathcal{R}_x$ is a subgroup of Z so if d is the least positive integer which appears in any of the groups $\mathcal{R}_x$ then there is some $V \subseteq U$ so that $\mathcal{R}|_V \cong d \cdot Z|_V$ and we get the exact sequence:

$$0 \rightarrow Z_V \rightarrow \mathcal{R} \rightarrow \mathcal{R}/Z_V \rightarrow 0$$

The sheaf $\mathcal{R}/Z_V$ is supported on the closed subset $(U - V)^-$ of X which has dimension $< n$ since X is irreducible. Therefore by the inductive hypothesis $H^p(X, Z) = 0$ for $p \geq n$. Therefore by the long exact sequence of cohomology we only need to show vanishing for Z_V . Let $Y = X - V$ and consider

$$0 \rightarrow Z_V \rightarrow Z \rightarrow Z_Y \rightarrow 0$$

and note that $\dim Y < \dim X$ since X is irreducible. Therefore $H^p(X, Z_Y) = 0$ for $p \geq n$ by the induction hypothesis. On the other hand Z is flasque (check surjectivity) and therefore $H^p(X, Z) = 0$ for $p > 0$. Therefore by the long exact sequence of cohomology it follows that $H^p(X, Z_V) = 0$ for $p > n$ as desired. $\square$

Example 6.11 (Čech Cohomology of $\mathcal{O}(n)$ on $\mathbb{P}^1$). We will compute the cohomology of the twisting sheaves on $\mathbb{P}^1$ using the standard affine cover. Let $U_1 = \operatorname{Spec} k[t]$ and $U_2 = \operatorname{Spec} k[u]$ with $u = t^{-1}$ on the intersection $U_{12} = \operatorname{Spec} k[t, t^{-1}]$. The bundle $\mathcal{O}(n)$ is defined by the transition function t^{-n} on the overlap. This means that a section s_2 on U_2 corresponds to the function $t^n s_2(t^{-1})$ when viewed on U_{12} .

The Čech complex is:

$$C^0(\mathcal{O}(n)) \xrightarrow{d^0} C^1(\mathcal{O}(n)) \xrightarrow{d^1} 0$$

Explicitly, this maps a pair of sections $(P(t), Q(u)) \in k[t] \oplus k[u]$ to:

$$d^0(P, Q) = P(t) - t^n Q(t^{-1}) \in k[t, t^{-1}]$$

Computing H^0 : The kernel consists of pairs where $P(t) = t^n Q(t^{-1})$. If $P(t) = \sum a_i t^i$ and $Q(u) = \sum b_j u^j$, equality implies:

$$\sum_{i=0}^{\infty} a_i t^i = \sum_{j=0}^{\infty} b_j t^{n-j}$$

Comparing coefficients, we require $b_j = a_{n-j}$.

- If $n < 0$, the powers of t on the RHS are all negative (since $j \geq 0$), while LHS powers are non-negative. No non-zero solution exists.
- If $n \geq 0$, valid indices are $0 \leq j \leq n$, corresponding to $0 \leq i \leq n$. This gives $n + 1$ independent choices $(1, t, \dots, t^n)$.

Thus $h^0(\mathbb{P}^1, \mathcal{O}(n)) = \max(0, n + 1)$.

Computing H^1 : The first cohomology group is the cokernel of d^0 . Recall that $\{t^k \mid k \in \mathbb{Z}\}$ is a basis for $k[t, t^{-1}]$. The image of d^0 contains all t^k for $k \geq 0$ (from $P(t)$) and all t^k for $k \leq n$ (from $t^n Q(t^{-1})$).

- If $n \geq -1$, the union of these sets is all integers k , so the map is surjective and $H^1(\mathbb{P}^1, \mathcal{O}(n)) = 0$.
- If $n < -1$, the integers strictly between n and 0 are not in the image. The missing monomials are $t^{n+1}, t^{n+2}, \dots, t^{-1}$. The number of such monomials is $(-1) - (n + 1) + 1 = -n - 1$.

Thus $h^1(\mathbb{P}^1, \mathcal{O}(n)) = \max(0, -n - 1)$.

One should verify that the "residue pairing"

$$\langle f, g \rangle = \operatorname{Coeff}_{t^{-1}}(f(t) \cdot g(t))$$

gives a perfect pairing

$$H^0(\mathbb{P}^1, \mathcal{O}(n)) \times H^1(\mathbb{P}^1, \mathcal{O}(-n-2)) \rightarrow k$$

which is an example of Serre Duality (Theorem 7.2).

Definition 6.12 (Euler Characteristic). Suppose that X is a Noetherian scheme of finite dimension. The *Euler characteristic* is the alternating sum of the dimensions of the cohomology groups

$$\chi(X, \mathcal{F}) = \sum_{p=0}^{\infty} (-1)^p h^p(X, \mathcal{F})$$

which is well defined by Theorem 6.10.

This definition should be thought of as somewhat analogous to the famous equation for a three dimensional polyhedron

$$\chi = V - E + F$$

which states that the Euler characteristic equals the number of vertices minus edges plus faces. This reduces a very complicated topological idea to a matter of counting.

Theorem 6.13 (Additivity of Euler Characteristic). *If*

$$0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$$

is an exact sequence of sheaves then:

$$\chi(\mathcal{F}_2) = \chi(\mathcal{F}_1) + \chi(\mathcal{F}_3)$$

Proof. Take the long exact sequence of cohomology

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(\mathcal{F}_1) & & H^1(\mathcal{F}_1) & & H^2(\mathcal{F}_1) & & \cdots \\ & & \downarrow & \nearrow & \downarrow & \nearrow & \downarrow & \nearrow & \\ & & H^0(\mathcal{F}_2) & & H^1(\mathcal{F}_2) & & H^2(\mathcal{F}_2) & & \cdots \\ & & \downarrow & \nearrow & \downarrow & \nearrow & \downarrow & \nearrow & \\ & & H^0(\mathcal{F}_3) & & H^1(\mathcal{F}_3) & & H^2(\mathcal{F}_3) & & \cdots \end{array}$$

and note since $\dim_k$ is additive the alternating sum of the dimensions of the homologies along this sequence is 0. This sum is exactly the alternating sum of the Euler characteristics as desired. $\square$

Definition 6.14 (Genus). The genus is defined as the dimension of the first cohomology group of the structure sheaf:

$$g = h^1(X, \mathcal{O}_X)$$

In general, the genus says something about the number of holes a topological space has. For instance a sphere has genus 0 but a torus has genus 1.

Remark 6.15. In the case of a smooth proper curve

$$\chi(X, \mathcal{F}) = h^0(X, \mathcal{F}) - h^1(X, \mathcal{F})$$

by Theorem 6.10.

7. SERRE DUALITY AND THE RIEMANN-ROCH THEOREM

Finally, we can prove the Riemann-Roch theorem. As before let C be a smooth proper curve.

Theorem 7.1 (Cheap Riemann-Roch). *If L is a line bundle over C then*

$$h^0(L) - h^1(L) = \deg L + 1 - g$$

where $h^p(L) := h^p(C, L)$.

Proof. Consider the line bundle $L_1 \otimes L_2$ where $L_2 = \mathcal{L}^D$ for some effective divisor D . Now consider the exact sequence in Theorem 4.10

$$0 \rightarrow L_1 \rightarrow L_1 \otimes L_2 \rightarrow S \rightarrow 0 \quad S := \prod_i k(P_i)^{a_i}$$

Using Theorem 6.13 we obtain:

$$\chi(L_1 \otimes L_2) = \chi(L_1) + \chi(S)$$

However note that $h^0(S) = \deg L_2$ by construction and $h^p(S) = 0$ for $p > 0$ since S is flasque. Therefore $\chi(S) = \deg L_2$ and we obtain

$$(1) \quad \chi(L_1 \otimes L_2) = \chi(L_1) + \deg L_2$$

Now take a "common ancestor" L' of L and $\mathcal{O}_C$ with

$$L = L' \otimes \mathcal{L}^{D_1} \quad \mathcal{O}_C = L' \otimes \mathcal{L}^{D_2}$$

for effective divisors D_1 and D_2 . By applying Eq. (1) to L and $\mathcal{O}_C$ and comparing it follows that

$$\chi(L) = \chi(\mathcal{O}_C) + \deg L$$

which is the statement that the Euler characteristic is additive in degree. Now use Remark 6.15 to rewrite the LHS and RHS as

$$h^0(L) - h^1(L) \quad h^0(\mathcal{O}_C) - h^1(\mathcal{O}_C) + \deg L = \deg L + 1 - g$$

respectively, using definitions and $h^0(\mathcal{O}_C) = 1$. This is the desired equation. $\square$

You may notice that the canonical divisor does not appear in this theorem. This is because we have not yet proven the full theorem. In order to achieve the desired form and draw connections to the canonical divisor, we need to invoke an extremely deep result known as *Serre duality*. I will recount a proof sketch of Serre duality due to Ngô Bảo Châu. A full proof may be found in [1] or [10].

Theorem 7.2 (Serre Duality). *For every line bundle L over C there is a natural perfect pairing:*

$$\langle \cdot, \cdot \rangle : H^0(C, L) \times H^1(C, \Omega_C \otimes L^\vee) \rightarrow k$$

Proof. The proof is quite involved, so we give a sketch of the overarching logical steps. These are verifying the theorem for $\mathbb{P}^1$, reducing a general curve to $\mathbb{P}^1$ via a finite morphism, and constructing the "trace map" to define the duality on the curve.

Step 1: The Case $C = \mathbb{P}^1$: Recall from Example 6.11 there is a perfect pairing:

$$H^0(\mathbb{P}^1, \mathcal{O}(n)) \times H^1(\mathbb{P}^1, \mathcal{O}(-n-2)) \rightarrow k$$

Therefore to show Serre duality for $\mathbb{P}^1$ we only need to prove that $\Omega_{\mathbb{P}^1} = \mathcal{O}(-2)$. Identifying $\mathbb{P}^1$ as $\text{Spec } k[t] \cup \text{Spec } k[u]$ and differentiating the transition function $t \mapsto u^{-1}$ gives $dt \mapsto -u^{-2}du$ which corresponds to the definition of $\mathcal{O}(-2)$.

Step 2: Reduction via Finite Morphism: For any affine open U of C , the existence of a finite morphism $f : U \rightarrow \mathbb{A}^1$ follows directly from Noether Normalization. We are able to extend this to a map $f : C \rightarrow \mathbb{P}^1$ due to the properness of C , with all the points in $C \setminus U$ sent to $\infty \in \mathbb{P}^1$.

Now let L be a line bundle on C . Since f is a finite affine morphism, the cohomology of L on C is isomorphic to the cohomology of its pushforward f_*L on $\mathbb{P}^1$ (computation of Čech cohomology commutes with affine pushforwards).

Since f_*L is a vector bundle on $\mathbb{P}^1$, every vector bundle on $\mathbb{P}^1$ splits into a sum of line bundles (Grothendieck's Theorem), and we know duality holds for line bundles on $\mathbb{P}^1$, duality must hold for f_*L . This gives us a perfect pairing:

$$H^0(\mathbb{P}^1, f_*L) \times H^1(\mathbb{P}^1, (f_*L)^\vee \otimes \Omega_{\mathbb{P}^1}) \rightarrow k$$

Since $H^0(\mathbb{P}^1, f_*L) \cong H^0(C, L)$, we have found *some* dual space. The difficulty lies in identifying the complicated object on the right side with the sheaf of differentials on C . Specifically, we must show an isomorphism of sheaves:

$$(f_*L)^\vee \otimes \Omega_{\mathbb{P}^1} \cong f_*(L^\vee \otimes \Omega_C)$$

Step 3: The Trace and The Different: The final and most difficult step is to identify the abstract dual space from Step 2 with the concrete sheaf of differentials Ω_C . We have established a dual pairing on $\mathbb{P}^1$, which lifts to a dual pairing on C . We must show that the object defining this lifted pairing is isomorphic to Ω_C .

Since $f : C \rightarrow \mathbb{P}^1$ is a finite morphism of degree n , locally the ring of functions B on C is a free module of rank n over the ring of functions A on $\mathbb{P}^1$. For any function $g \in B$, multiplication by g defines a linear map $M_g : B \rightarrow B$. We define the trace of g as the trace of this linear transformation matrix:

$$\text{tr}_{C/\mathbb{P}^1}(g) := \text{Trace}(M_g)$$

While g lives on the curve C , $\text{tr}(g)$ is a regular function on the line $\mathbb{P}^1$.

We can now define a pairing $\langle a, b \rangle = \text{tr}(a \cdot b)$. We seek the "Algebraic Dual" sheaf, which consists of all rational functions $y \in K(C)$ such that the pairing with any regular function lands in $\mathcal{O}_{\mathbb{P}^1}$:

$$\mathcal{C} = \{y \in K(C) \mid \text{tr}(y \cdot \mathcal{O}_C) \subseteq \mathcal{O}_{\mathbb{P}^1}\}$$

If the map were unramified, $\mathcal{O}_C$ would be its own dual. However, at ramification points, the trace "overcounts". To make the pairing perfect, the dual $\mathcal{C}$ must be larger than $\mathcal{O}_C$, it must allow specific poles. This defines a divisor $\mathfrak{D}$ called the *Different*, such that the algebraic dual is the inverse ideal sheaf $\mathfrak{D}^{-1}$.

Now we look at the differentials. We compare Ω_C to the pullback $f^*\Omega_{\mathbb{P}^1}$. At a ramification point, the map "folds over," causing the pullback differential to vanish relative to the curve's differential. Therefore, Ω_C is "larger"

than the pullback (it generates the module, while the pullback is a submodule). The divisor describing this gap is the *Ramification Divisor* R .

$$\Omega_C \cong f^* \Omega_{\mathbb{P}^1} \otimes \mathcal{L}^R$$

To prove the theorem, we must show the Algebraic Dual ($\mathfrak{D}$) matches the Geometric Dual (R). We work with the completions of the local rings at a ramification point P mapping to Q . Let $A = \hat{\mathcal{O}}_Q$ be the local ring downstairs (with uniformizer t) and $B = \hat{\mathcal{O}}_P$ be the local ring upstairs (with uniformizer u). Since the extension is totally ramified, the structure theorem for principal ideal domains implies B is generated over A by u , which satisfies an *Eisenstein polynomial* $P(z) \in A[z]$:

$$P(z) = z^e + a_{e-1}z^{e-1} + \cdots + a_0 = 0$$

where a_i are divisible by t , and a_0 has valuation 1 (divisible by t but not t^2).

- **Algebraic (Euler's Formula):** A standard result in local algebra states that the dual of B with respect to the trace is generated by the inverse of the derivative $P'(u)$.

$$\text{Generator}(\mathfrak{D}^{-1}) = \frac{1}{P'(u)}$$

- **Geometric:** We compare differentials. Since $P(u) = 0$, we differentiate to get $P'(u)du + (\text{terms involving } dt) = 0$. This implies the relation $f^*dt \cong P'(u)du$. Solving for the generator du , we see it differs from the base f^*dt by exactly $\frac{1}{P'(u)}$.

Since P is Eisenstein, the valuation of $P'(u)$ is dominated by the leading term eu^{e-1} (the lower terms $a_k z^k$ have coefficients divisible by t , making them higher order). Thus, both the algebraic generator and the geometric generator differ from the base by exactly the same factor. This implies $\mathfrak{D} = R$, establishing the isomorphism:

$$(f_* \mathcal{O}_C)^\vee \otimes \Omega_{\mathbb{P}^1} \cong f_*(\Omega_C)$$

This allows us to transport the Serre Duality pairing from $\mathbb{P}^1$ to C via the trace, completing the proof. $\square$

Remark 7.3 (Integration). While the proof of Theorem 7.2 utilizes the algebraic trace, Serre Duality has a natural interpretation in terms of integration. Over the complex numbers, elements of $H^1(C, \mathcal{O}_C)$ can be identified with $(0, 1)$ -forms η , and elements of $H^0(C, \Omega_C)$ are holomorphic $(1, 0)$ -forms ω . The pairing is given by the integral of their wedge product over the curve:

$$\langle \omega, \eta \rangle = \int_C \omega \wedge \eta \in \mathbb{C}$$

In the algebraic setting where we use Čech cohomology, this integration corresponds to the *sum of residues*. If $\xi \in H^1(\mathcal{O}_C)$ is represented by local data (f_{ij}) on overlaps, and ω is a global differential, their product defines a differential on the intersections. The duality pairing is obtained by taking the sum of the residues of these local differentials. The "trace" map in the proof above can be viewed as the algebraic generalization of this residue operation.

Theorem 7.4 (Riemann-Roch Theorem). *Suppose that L is a line bundle over C . Then*

$$h^0(L) - h^0(\Omega_C \otimes L^\vee) = \deg L + 1 - g$$

and $g = h^0(\Omega_C)$.

Proof. By Theorem 7.1:

$$(2) \quad h^0(L) - h^1(L) = \deg L + 1 - g$$

Note that a perfect pairing of vector spaces means that they have the same dimension. Therefore by Theorem 7.2 we can write

$$h^1(L) = h^0(\Omega_C \otimes L^\vee)$$

and plugging into Eq. (2) yields the Riemann-Roch theorem. $\square$

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